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JEE MAIN PHYSICS DPP: VELOCITY-TIME GRAPHS

Topic: v-t **Graphs, Area Under Curve & Graphical Derivation of Motion Equations** | **Time: 45 Min** | **Max Marks: 80**

Marking Scheme: **+4, -1** | Designed & Curated by: **Syed Mazhar Ali**

Q1. The slope of the tangent to a velocity-time (v-t) graph at any given instant gives:

(a) Instantaneous velocity
(b) Instantaneous acceleration
(c) Total displacement
(d) Jerk

Q2. The area bounded by a velocity-time graph and the time axis between instants t_1 and t_2 represents:

(a) Total distance only
(b) Instantaneous acceleration
(c) Net displacement
(d) Average velocity

Q3. A particle moves along a straight line. Its v-t graph consists of a triangle above the time axis with base 0 to 4 s and height 6 m/s, and a triangle below the axis with base 4 to 6 s and depth -4 m/s. The total distance travelled from $t = 0$ to 6 s is:

(a) 8 m
(b) 12 m
(c) 4 m
(d) 16 m

Q4. In the previous question, the net displacement of the particle in the interval $t = 0$ to 6 s is:

(a) 8 m
(b) 16 m
(c) 12 m
(d) 4 m

Q5. In the graphical derivation of $s = ut + \frac{1}{2}at^2$, the v-t graph is a straight line from $(0, u)$ to (t, v). The total area under the graph is partitioned into:

(a) Two equal rectangles
(b) A rectangle of area ut and a triangle of area $\frac{1}{2}(v - u)t$
(c) A triangle of area ut and a square of area $\frac{1}{2}at^2$
(d) A trapezium of area $(u + v)t$

Q6. A car accelerates from rest at a constant rate α for time t_1 and then retards at a constant rate β for time t_2 to come to rest. The ratio $\dfrac{t_1}{t_2}$ in terms of α and β is:

(a) $\dfrac{\alpha}{\beta}$
(b) $\sqrt{\dfrac{\beta}{\alpha}}$
(c) $\dfrac{\beta}{\alpha}$
(d) $\dfrac{\alpha^2}{\beta^2}$

Q7. A particle starts from rest and its v-t graph is a semi-circle of radius $R = 4$ units on the upper half-plane with diameter lying along the time axis from $t = 0$ to $t = 8$ s. The total displacement is:

(a) 16π m
(b) 4π m
(c) 32π m
(d) 8π m

Q8. The v-t graph of an object moving on a line is a trapezium with parallel sides of lengths T (total time) and t_0 (constant speed interval), and maximum height v_0. The total displacement is:

(a) $\dfrac{1}{2}(T + t_0)v_0$
(b) $(T + t_0)v_0$
(c) $\dfrac{1}{2}(T - t_0)v_0$
(d) $\dfrac{Tt_0v_0}{T + t_0}$

Q9. Which of the following velocity-time graphs represents a ball thrown vertically upwards and caught upon returning? (Taking upward as positive):

(a) A horizontal line above the time axis
(b) A straight line with negative slope crossing the time axis
(c) A straight line with positive slope starting from origin
(d) A parabola opening downwards

Q10. The velocity-time relation of a body is represented by $v(t) = 3t^2 - 6t$ m/s. The net displacement from $t = 0$ to $t = 3$ s obtained by integration (area under curve) is:

(a) 9 m
(b) 4 m
(c) 0 m
(d) -4 m

Q11. In the previous question, the total distance travelled by the particle from $t = 0$ to $t = 3$ s is:

(a) 0 m
(b) 4 m
(c) 5 m
(d) 8 m

Q12. The v-t graph of a particle is shown as an isosceles triangle with base along the time axis from $t = 0$ to $t = 10$ s and peak velocity $v_{\max} = 20$ m/s at $t = 5$ s. The average acceleration from $t = 0$ to $t = 5$ s is:

(a) $+4$ m/s^2
(b) $+2$ m/s^2
(c) -4 m/s^2
(d) 0 m/s^2

Q13. For the same motion as in Q12, the average velocity of the particle over the entire 10 s interval is:

(a) 20 m/s
(b) 10 m/s
(c) 5 m/s
(d) 15 m/s

Q14. A body moves in a straight line such that its v-t curve is given by $v = \alpha\sqrt{t}$. The distance travelled by the body in the first 4 seconds is:

(a) $8\alpha/3$
(b) 4α
(c) $16\alpha/3$
(d) 2α

Q15. In a v-t graph, if the curve is convex downwards ($\frac{d^2v}{dt^2} > 0$), it implies that:

(a) Acceleration is constant
(b) Acceleration is decreasing with time
(c) Velocity is constant
(d) Acceleration is increasing with time

Q16. A car and a truck start from rest at $t = 0$ from the same point. The car moves with constant acceleration $a_c = 4$ m/s^2, and the truck moves with constant velocity $v_t = 16$ m/s. From their v-t graphs, the time when the car overtakes the truck is:

(a) 8 s
(b) 4 s
(c) 12 s
(d) 16 s

Q17. In Q16, the maximum separation between the truck and the car occurs at time t equal to:

(a) 2 s
(b) 4 s
(c) 8 s
(d) 6 s

Q18. If the v-t graph of a particle is a straight line parallel to the time axis at $v = v_0 \neq 0$, the corresponding acceleration-time (a-t) graph is:

(a) A straight line parallel to time axis at $a = v_0$
(b) A line with positive slope
(c) The time axis itself ($a = 0$)
(d) A vertical straight line

Q19. A ball is dropped from height H onto a floor and bounces back elastically to the same height. Assuming negligible contact time, the v-t graph consists of:

(a) A continuous sine wave
(b) Alternating horizontal steps
(c) Triangular spikes of same sign
(d) Repeated parallel line segments with negative slope $-g$ and vertical jumps at impacts

Q20. A body moves with uniform acceleration. In a v-t graph, the area of the trapezium between $t = 0$ and $t = t$ is Area $= \frac{1}{2}(u + v)t$. Substituting $t = \dfrac{v-u}{a}$ directly derives which kinematic equation?

(a) $v^2 = u^2 + 2as$
(b) $s = ut + \frac{1}{2}at^2$
(c) $v = u + at$
(d) $s_n = u + \frac{a}{2}(2n - 1)$

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(b)	**6**	(c)	**11**	(d)	**16**	(a)
2	(c)	**7**	(d)	**12**	(a)	**17**	(b)
3	(d)	**8**	(a)	**13**	(b)	**18**	(c)
4	(a)	**9**	(b)	**14**	(c)	**19**	(d)
5	(b)	**10**	(c)	**15**	(d)	**20**	(a)

Detailed Solutions

Sol 1. Correct Option: (b)
By definition, the derivative of velocity with respect to time gives instantaneous acceleration:

$$\text{Slope} = \frac{dv}{dt} = a_{\text{inst}}$$

Sol 2. Correct Option: (c)
The definite integral of velocity with respect to time represents net displacement:

$$\Delta x = \int_{t_1}^{t_2} v(t)dt = \text{Net Area under } v\text{-}t \text{ graph}$$

Areas below the time axis are subtracted algebraically.

Sol 3. Correct Option: (d)
- Area above axis (0 to 4 s): $A_1 = \frac{1}{2} \times 4 \times 6 = 12$ m.
- Area below axis (4 to 6 s): $|A_2| = \frac{1}{2} \times 2 \times |-4| = 4$ m.

Total distance $= A_1 + |A_2| = 12 + 4 = 16$ m.

Sol 4. Correct Option: (a)
Net displacement takes signed areas into account:

$$\Delta x = A_1 - |A_2| = 12 - 4 = 8 \text{ m}$$

Sol 5. Correct Option: (b)
The trapezium under the v-t graph from $(0, u)$ to (t, v) splits into:
- Base rectangle: $\text{Area}_1 = u \times t = ut$
- Upper triangle: $\text{Area}_2 = \frac{1}{2} \times t \times (v - u) = \frac{1}{2}at^2$

Total area: $s = ut + \frac{1}{2}at^2$.

Sol 6. Correct Option: (c)
Peak velocity attained: $v_{\text{max}} = \alpha t_1 = \beta t_2$.
Rearranging gives:

$$\frac{t_1}{t_2} = \frac{\beta}{\alpha}$$

Sol 7. Correct Option: (d)
Displacement is the area of the semicircle of radius $R = 4$ units:

$$\text{Area} = \frac{1}{2}\pi R^2 = \frac{1}{2}\pi(4)^2 = 8\pi \text{ m}$$

Sol 8. Correct Option: (a)
From standard trapezium geometry:
Displacement $s = \frac{1}{2}$(sum of parallel sides) $\times$ (height):

$$s = \frac{1}{2}(T + t_0)v_0$$

Sol 9. Correct Option: (b)
For vertical projectile motion, acceleration is constant ($a = -g$). The velocity-time relation is $v(t) = u - gt$, which is a straight line of negative slope $-g$, crossing $v = 0$ at maximum height $t = u/g$.

Sol 10. Correct Option: (c)
Displacement by integration:

$$\Delta x = \int_0^3 (3t^2 - 6t)dt = \left[t^3 - 3t^2\right]_0^3 = 0 \text{ m}$$

Sol 11. Correct Option: (d)
Velocity crosses zero at $v = 3t(t-2) = 0 \implies t = 2$ s.
- Displacement in $[0, 2 \text{ s}]$: $\left[t^3 - 3t^2\right]_0^2 = 8 - 12 = -4$ m.
- Displacement in $[2, 3 \text{ s}]$: $(27 - 27) - (8 - 12) = +4$ m.

Total distance $= |-4| + |+4| = 8$ m.

Sol 12. Correct Option: (a)
Average acceleration is the slope of the secant line:

$$a_{\text{avg}} = \frac{v(5) - v(0)}{5 - 0} = \frac{20 - 0}{5} = +4 \text{ m/s}^2$$

Sol 13. Correct Option: (b)
Total displacement = Area of triangle:

$$s = \frac{1}{2} \times 10 \times 20 = 100 \text{ m}$$

Average velocity:

$$v_{\text{avg}} = \frac{s}{\Delta t} = \frac{100 \text{ m}}{10 \text{ s}} = 10 \text{ m/s}$$

Sol 14. Correct Option: (c)
Displacement is the area under $v(t) = \alpha t^{1/2}$ from $t = 0$ to 4 s:

$$s = \int_0^4 \alpha t^{1/2} dt = \alpha \left[\frac{2}{3} t^{3/2}\right]_0^4$$

$$= \frac{2}{3}\alpha(8) = \frac{16\alpha}{3}$$

Sol 15. Correct Option: (d)
$\frac{d^2v}{dt^2} = \frac{d}{dt}\left(\frac{dv}{dt}\right) = \frac{da}{dt} > 0.$
Since the time derivative of acceleration is positive, acceleration is increasing with time.

Sol 16. Correct Option: (a)
For the car to overtake the truck, displacements must be equal ($s_c = s_t$):
- Truck: $s_t = v_t t = 16t$
- Car: $s_c = \frac{1}{2}a_c t^2 = \frac{1}{2}(4)t^2 = 2t^2$

$2t^2 = 16t \implies t(2t - 16) = 0 \implies t = 8$ s.

Sol 17. Correct Option: (b)
Separation between them is maximized when their relative velocity becomes zero ($v_c = v_t$):

$$a_c t = v_t \implies 4t = 16 \implies t = 4 \text{ s}$$

Sol 18. Correct Option: (c)
Since $v = v_0 =$ constant, the slope $\frac{dv}{dt} = 0$ for all t. Thus, the acceleration is zero everywhere, which is represented by the time axis line itself.

Sol 19. Correct Option: (d)
During each free fall and rise, slope is constant $\frac{dv}{dt} = -g$. At each elastic bounce, the velocity abruptly jumps from $-v_0$ to $+v_0$ instantaneously, resulting in repeated parallel lines with vertical discontinuities.

Sol 20. Correct Option: (a)
Area of trapezium:

$$s = \frac{u+v}{2}t$$

Substituting $t = \frac{v-u}{a}$:

$$s = \left(\frac{v+u}{2}\right)\left(\frac{v-u}{a}\right) = \frac{v^2 - u^2}{2a}$$

$$v^2 = u^2 + 2as$$